

very careful with the manufacturing process. The smallest of problems can lead to very disastrous PCBs." The boards need to be cleaned thoroughly and must be checked and re-checked for any remaining particles post processing before you add any components. He explains, "You cannot leave any flux on your boards post cleaning. Unless the board is clean of every individual particle, you are going to have problems."

Through-hole is an area that sees a lot of problems. However, are there solutions that can take care of these. Improved plating distribution has had positive effects like improving the reliability of the PTHs. Electrolytic plating is a cost-effective method for doing this. It uniformly increases the thickness of circuits and results in improved tracks.

For improved thermal shock and cycling, however, acid copper plating process is the one to go for. Throwing power can also be increased by improvements in plating cell design with anode configurations and proper selection of circuit components. Pulse plating could also help in this regard. Ilmer says, "There was once a case of phenolic paper boards being used for PCBs."

The material for the board is also important. Phenolic paper boards help in cutting cost, but bring in their own set of problems. Compared to glass epoxy boards, these have less adhesion for copper cladding and are more prone to environmental degradation and chemical attacks.

These are also mechanically weaker, which facilitates easy drilling and shaping. However, this ease can be easily looked over as these

Factors at a glance

Robert Tarzwell and Ken Bahl have explained some factors that affect the reliability of PCBs over a long term, in a paper they published some time ago:

1. The more assembly cycles that a circuit experiences, the less reliable the product will be in the field
2. The hotter or longer the assembly cycles are, the less reliable the product will be
3. The thicker the laminates, the less reliable the product will be, by a considerable factor
4. The lower the T/g, the less reliable the board
5. Copper ductility of the plating bath greatly affects reliability
6. Copper plating thickness, if too thin, will drastically lower reliability
7. The cure and selection of prepreg in multilayers plays a big role in future life cycles
8. Drilling quality affects hole-plating quality and therefore reduces the life in the field
9. The de-smear process is critical to reduce via cracking
10. Any small pinhole in the copper wall is a concern for reliability
11. A high number of thermal cycles over 100°C delta causes reduced operational life

happen to be more brittle than glass epoxy boards. This highlights the fact that the material used in the boards is also a matter of concern.

Laminate materials present their own set of problems. These can be brittle and cause problems during inner layer treatment, leading to reduced bond strength between multiple layers of the board. Minute factors like moisture absorption become important factors at this point.

To ensure improved bond strength, longer lamination cycle times and pre-lamination bake cycles on inner layer materials become important as well. The pre-heat must be set right for the process to go smoothly.

Also, laminate materials could lead to poor plating adhesion due to the fillers involved. This calls for de-smearing the resin prior to plating.

Recently, due to the influx of wearables, there is a high demand for flexible PCBs. Rigid and flex PCBs used in designing often use different materials, with the rigid sections consisting of more layers than the flexible ones. To improve

reliability with additional layers of flexible material, stiffeners are added to bring rigidity to the PCBs. These are usually made of stainless steel or aluminium, mixed with a dielectric material. A common practice to avoid cracking at the convergence of rigid and flexible areas is to not place components in those locations.

A good PCB is a fine line indeed

Therefore you cannot depend solely on the EDA tool to get it right; you must maintain your own personal library for reliable components. Once a PCB design is with the manufacturer, you must recheck for the right components. Deshpande says, "Once I sent a design to a manufacturer and got back a list of components that my components could be replaced with. This helped with the final product." But you cannot always rely on things going your way, can you?

You can neither let the PCB stay in the flux for too long, nor can you go with just about any board. Everything must work just right until you have a working PCB that you can rely on. Constant vigilance seems to be the key to improving reliability. The age-old proverb 'Toiling in the Sun' will eventually get you a PCB that will not blow up in your hands. **EFY**

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